Lecture 23

# Discrete series and the rank one unitary picture

**Objectives.** *We realise the holomorphic discrete series of* $\mathrm{SL}(2,\mathbb{R})$ *on weighted spaces of holomorphic functions on the disk. We then compare it with the finite-dimensional and principal series modules, and a map* $U_n$ *identifies its polynomials with a half-ladder of the principal series. Finally we state the full list of irreducible unitary representations, and we prove that a finite-dimensional unitary representation is trivial.*

By the end of this lecture you will be able to:

- Prove that $\Pi_n$ is an irreducible unitary representation on the weighted disk space $\mathcal{A}_n$ for each integer $n \geq 2$.
- Compute its $K$-types $n, n{+}2, \dots$ and the action of $H', E, F$, and identify its polynomials with a half-ladder of the principal series by the map $U_n$.
- Decide which lowest weights $\lambda$ occur in irreducible unitary representations of $\mathrm{SL}(2,\mathbb{R})$, and when such a representation factors through $\mathrm{PSL}(2,\mathbb{R})$.
- State Bargmann's classification, derive the positivity bounds behind it, and prove that every finite-dimensional continuous unitary representation of $\mathrm{SL}(2,\mathbb{R})$ is trivial.

## 23.1 The disk model

Lecture 22 found half-ladders at $\nu = n-1$. For $n \geq 2$ the $L^2$ norm of Lecture 22 is not invariant there, so a half-ladder needs a new norm. A half-ladder has one sign of $K$-type. Power series in $z$ have the same shape, because rotation multiplies $z^m$ by a power of $e^{2i\theta}$. So we look for holomorphic functions on a disk.

Let $G = \mathrm{SL}(2,\mathbb{R})$. We keep the conventions of Lecture 21:

$$W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k_\theta = \exp(\theta W) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}, \qquad K = \{k_\theta\}.$$

Thus $k_{2\pi} = I$ and $k_\pi = -I$. With $H_0 = \mathrm{diag}(1,-1)$ and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ put

$$H' = -iW, \qquad E = \tfrac{1}{2}(H_0 + iS), \qquad F = \tfrac{1}{2}(H_0 - iS) = \overline{E}.$$

Then $[H', E] = 2E$, $[H', F] = -2F$ and $[E, F] = H'$. In a representation $\Pi$, a vector $v$ has *$K$-type $k$* if $\Pi(k_\theta)v = e^{ik\theta}v$ for all $\theta$. If such a $v$ is smooth, then $d\Pi(H')v = kv$.

The disk carries an action of

$$\mathrm{SU}(1,1) = \left\{ \begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix} : |\alpha|^2 - |\beta|^2 = 1 \right\}.$$

Fix the Cayley matrix $\gamma = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ and set $\varphi(g) = \gamma g \gamma^{-1}$. For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$, multiplying out gives

$$\varphi(g) = \begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix}, \qquad \alpha = \tfrac{1}{2}\big(a + d + i(c - b)\big), \qquad \beta = \tfrac{1}{2}\big(a - d + i(b + c)\big). \tag{23.1}$$

Then $|\alpha|^2 - |\beta|^2 = ad - bc = 1$. Solving (23.1) for $a, b, c, d$ shows that $\varphi$ is an isomorphism of Lie groups from $G$ onto $\mathrm{SU}(1,1)$. Also

$$\varphi(k_\theta) = \mathrm{diag}(e^{-i\theta}, e^{i\theta}), \qquad \varphi(-I) = -I. \tag{23.2}$$

Conjugation by $\gamma$ is complex linear on all $2 \times 2$ matrices. A direct multiplication gives

$$\gamma H' \gamma^{-1} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \gamma E \gamma^{-1} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad \gamma F \gamma^{-1} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{23.3}$$

The matrix $\bar{\gamma}$, which is the matrix $c$ of Exercise 20.1, gives the triple in the order of Lecture 21. Our $\gamma$ reverses it, so that $\Pi_n$ below has positive $K$-types.

A function on an open subset of the complex plane is *holomorphic* if it is complex differentiable at every point.

We import three results from elementary complex analysis. The first is the local power-series theorem. A holomorphic function on a disk has a Taylor series converging uniformly on every smaller closed disk. The second concerns power series. A power series about the centre of a disk that converges at every point of the disk defines a holomorphic function there. The third follows from the Cauchy–Riemann equations. The real Jacobian of a holomorphic map is the squared modulus of its complex derivative. These are the analytic inputs used below. See MIT 18.112, the lectures on Cauchy's formula, power series and the Cauchy–Riemann equations.

Let $D = \{z \in \mathbb{C} : |z| < 1\}$ with closure $\overline{D}$. For $h \in \mathrm{SU}(1,1)$ as above and $z \in \overline{D}$ put

$$h \cdot z = \frac{\alpha z + \beta}{\bar{\beta} z + \bar{\alpha}}, \qquad j(h, z) = \bar{\beta} z + \bar{\alpha}.$$

**Lemma 23.1.** *Let $h, h_1, h_2 \in \mathrm{SU}(1,1)$ and $z \in \overline{D}$.*

*(a)* $|j(h,z)| \geq |\alpha| - |\beta| > 0$ *and* $|j(h,z)|^2 - |\alpha z + \beta|^2 = 1 - |z|^2$.

*(b)* $h$ *maps* $D$ *into* $D$ *and* $\overline{D}$ *into* $\overline{D}$, $(h_1 h_2) \cdot z = h_1 \cdot (h_2 \cdot z)$ *and* $j(h_1 h_2, z) = j(h_1, h_2 \cdot z)\, j(h_2, z)$.

*(c)* $d(h \cdot z)/dz = j(h,z)^{-2}$.

*Proof.* (a) Use $|z| \leq 1$ and $|\alpha|^2 = 1 + |\beta|^2$. Expanding both squares, the cross terms cancel and leave $(|\alpha|^2 - |\beta|^2)(1 - |z|^2)$.

(b) Dividing (a) by $|j(h,z)|^2$ gives

$$1 - |h \cdot z|^2 = \frac{1 - |z|^2}{|j(h,z)|^2}. \tag{23.4}$$

Hence $|h \cdot z| < 1$ whenever $|z| < 1$, so $h$ maps $D$ into $D$. Also, $1 - |h \cdot z|^2 \geq 0$ when $|z| \leq 1$, with equality exactly when $|z| = 1$. So $h$ maps $\overline{D}$ into $\overline{D}$.

Next,

$$j(h_1, h_2 \cdot z)\, j(h_2, z) = \bar{\beta}_1(\alpha_2 z + \beta_2) + \bar{\alpha}_1(\bar{\beta}_2 z + \bar{\alpha}_2).$$

The right side is the bottom row of $h_1 h_2$ applied to $(z,1)^T$, so it equals $j(h_1 h_2, z)$. In the same way, the top row of $h_1 h_2$ applied to $(z,1)^T$ is the numerator of $(h_1 h_2) \cdot z$. It equals $j(h_2, z)$ times the numerator of $h_1 \cdot (h_2 \cdot z)$. Dividing the two rows cancels $j(h_2, z)$ and gives $(h_1 h_2) \cdot z = h_1 \cdot (h_2 \cdot z)$.

(c) By the quotient rule the numerator is $\alpha(\bar{\beta} z + \bar{\alpha}) - \bar{\beta}(\alpha z + \beta) = 1$. ∎

Before we choose a norm, we look for an invariant area. By the identity (23.4) and Lemma 23.1(c), the area form $dA/(1 - |z|^2)^2$ on $D$ is invariant under $\mathrm{SU}(1,1)$. This is the *hyperbolic area*. The matrix $\bar{\gamma}$ moves the upper half-plane $\mathbb{H}$ of Lecture 20 to $D$, with $i$ going to 0, and this form becomes $\frac{1}{4} y^{-2} dx\, dy$. So it is a constant multiple of the invariant measure of Lecture 20, which is the left Haar measure of Lecture 16. Figure 23.1 shows this chart. Our $\gamma$ uses the lower half-plane instead, with $-i$ going to 0, so that the $K$-types of $\Pi_n$ are positive.

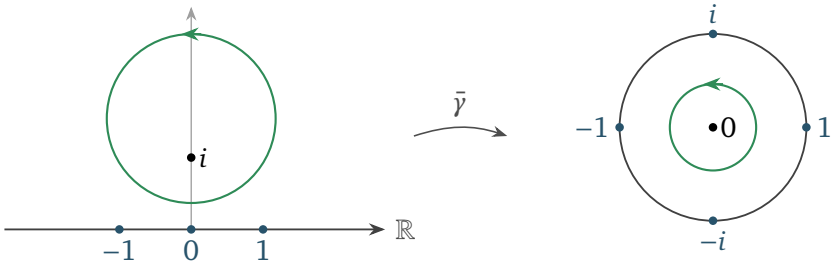



Figure 23.1: The chart $z \mapsto \bar{\gamma} \cdot z$. The points $-1$, 0 and 1 of the real line go to $i$, $-1$ and $-i$ on the boundary circle, and the point at infinity goes to 1. The point $i$ goes to 0, and the $K$-orbit of $ie$ goes to a centred circle. Arrows show $k_\theta$ for $\theta > 0$ in the parametrisation of this lecture.

Now take $f$ and $\Pi_n(h)f$ as in (23.5) below. At $z = h \cdot w$ the value $|\Pi_n(h)f|^2$ gains the factor $|j(h,w)|^{2n}$, and $(1 - |z|^2)^n$ gains the factor $|j(h,w)|^{-2n}$ by (23.4). These cancel. So the natural norm integrates $|f|^2(1 - |z|^2)^n$ against the hyperbolic area. This is why the power below is $n - 2$.

**Definition 23.2.** Let $n \geq 2$ be an integer and $dA = dx\, dy$. For $f$ holomorphic on $D$ put

$$\|f\|_n^2 = \frac{n-1}{\pi} \lim_{r \to 1} \int_{|z| \leq r} |f(z)|^2 (1 - |z|^2)^{n-2}\, dA(z) \in [0, \infty].$$

The *weighted disk space* $\mathcal{A}_n$ is the space of holomorphic $f$ on $D$ with $\|f\|_n < \infty$.

The limit exists because the integrand is nonnegative. For the same reason it equals the supremum of the integrals over compact regions in $D$ with smooth boundary.

**Lemma 23.3.** *If $f(z)=\sum_{m\ge 0} a_m z^m$ is holomorphic on $D$, then*

$$\|f\|_n^2=\sum_{m\ge 0} c_m|a_m|^2,\qquad c_m=\|z^m\|_n^2=\frac{m!\,(n-1)!}{(n+m-1)!}=\binom{n+m-1}{m}^{-1}.$$

*So $\mathcal{A}_n$ is a Hilbert space with orthonormal basis $e_m=c_m^{-1/2}z^m$. Moreover $\|g\|_n\le \sup_D|g|$ for bounded holomorphic $g$.*

*Proof.* On $|z|=s<1$ the series converges uniformly. Orthogonality of the functions $e^{im\theta}$ gives

$$\int_0^{2\pi}|f(se^{i\theta})|^2\,d\theta=2\pi\sum_m |a_m|^2 s^{2m}.$$

Integration in polar coordinates then gives

$$\frac{n-1}{\pi}\int_{|z|\le r}|f|^2(1-|z|^2)^{n-2}\,dA=\sum_m |a_m|^2\,(n-1)\int_0^r 2s^{2m+1}(1-s^2)^{n-2}\,ds.$$

The terms are nonnegative and increase with $r$, so we may let $r\to 1$ termwise. With $t=s^2$ and repeated integration by parts,

$$(n-1)\int_0^1 t^m(1-t)^{n-2}\,dt=(n-1)\frac{m!\,(n-2)!}{(n+m-1)!}=c_m.$$

Conversely, let $(a_m)_{m\ge 0}$ be complex numbers with $C=\sum_m c_m|a_m|^2<\infty$. Then $|a_m|^2\le C\binom{n+m-1}{m}\le C(m+1)^{n-1}$, so $\sum a_m z^m$ converges on $D$. By the power-series import, its sum $f$ is holomorphic on $D$. The formula above then gives $\|f\|_n^2=C$. Hence $f\mapsto (c_m^{1/2}a_m)_m$ is an isometry of $\mathcal{A}_n$ onto $\ell^2$. This gives completeness, the inner product by polarisation, and the basis. Finally $\|g\|_n^2\le \sup|g|^2\,\|1\|_n^2=\sup|g|^2$, since $c_0=1$. ∎

**Theorem 23.4.** *For $h\in \mathrm{SU}(1,1)$ and $f\in\mathcal{A}_n$ define*

$$(\Pi_n(h)f)(z)=j(h^{-1},z)^{-n}\,f(h^{-1}\cdot z). \tag{23.5}$$

*Then $g\mapsto \Pi_n(\varphi(g))$ is a strongly continuous unitary representation of $G$ on $\mathcal{A}_n$. We write $\Pi_n(g)$ for $\Pi_n(\varphi(g))$.*

*Proof. Homomorphism.* Apply the cocycle identity of Lemma 23.1(b) to $h_2^{-1}$ and $h_1^{-1}$:

$$(\Pi_n(h_1)\Pi_n(h_2)f)(z)=j(h_1^{-1},z)^{-n}\,j(h_2^{-1},h_1^{-1}\cdot z)^{-n}f(h_2^{-1}h_1^{-1}\cdot z)=(\Pi_n(h_1h_2)f)(z).$$

*Isometry.* Substitute $z=h\cdot w$. The cocycle identity with $h^{-1}h=I$ gives $j(h^{-1},h\cdot w)=j(h,w)^{-1}$. With the identity (23.4) for $1-|h\cdot w|^2$ and the derivative formula of Lemma 23.1(c) we get

$$\begin{aligned}&|(\Pi_n(h)f)(h\cdot w)|^2(1-|h\cdot w|^2)^{n-2}\left|\frac{d(h\cdot w)}{dw}\right|^2\\&\quad=|j(h,w)|^{2n}|f(w)|^2\frac{(1-|w|^2)^{n-2}}{|j(h,w)|^{2n-4}}\frac{1}{|j(h,w)|^4}.\end{aligned}$$

The powers of $|j(h,w)|$ cancel, leaving $|f(w)|^2(1-|w|^2)^{n-2}$.

For a holomorphic map the real Jacobian determinant is $|d(h\cdot w)/dw|^2$, by the Cauchy–Riemann equations. So on each compact region $R\subset D$, the change of variables formula equates the weighted integral of $|\Pi_n(h)f|^2$ over $h\cdot R$ with that of $|f|^2$ over $R$.

The map $w\mapsto h\cdot w$ is a bijection of $D$ with inverse $w\mapsto h^{-1}\cdot w$. So the regions $h\cdot R$ also run over all compact regions. Taking suprema gives $\|\Pi_n(h)f\|_n=\|f\|_n$. Thus $\Pi_n(h)$ is an isometry with inverse $\Pi_n(h^{-1})$, hence unitary.

*Continuity.* Let $f,g$ be polynomials. By the lower bound for $|j(h,z)|$ in Lemma 23.1(a), the function $\overline{g(z)}(\Pi_n(h)f)(z)(1-|z|^2)^{n-2}$ is continuous on $\mathrm{SU}(1,1)\times\overline{D}$. Here $n\geq 2$ is used at the boundary. It is uniformly continuous on $U\times\overline{D}$ for a compact neighbourhood $U$ of $I$. So $h\mapsto\langle g,\Pi_n(h)f\rangle_n$ is continuous. Polynomials are dense and $\|\Pi_n(h)\|=1$, so an $\varepsilon/3$ argument extends this to all $f,g$. Then

$$\|\Pi_n(h)f-f\|_n^2=2\|f\|_n^2-2\,\mathrm{Re}\langle f,\Pi_n(h)f\rangle_n\to 0\qquad(h\to I).$$

Unitarity moves this to every other point. ∎

**Remark 23.5.** Put $\delta=\mathrm{diag}(1,-1)$. Since $\bar\gamma=\gamma\delta$, the matrix $\bar\gamma$ in place of $\gamma$ gives $\Pi_n^-(g)=\Pi_n(\delta g\delta^{-1})$. Conjugation by $\delta$ preserves $G$, although $\delta\notin G$. Since $\delta k_\theta\delta^{-1}=k_{-\theta}$, the $K$-types of $\Pi_n^-$ are the negatives of those of $\Pi_n$. Which one is called holomorphic depends on the Cayley matrix and the sign of the $K$-angle. We label them by $K$-types.

## 23.2 $K$-types and Lie algebra operators

**Proposition 23.6.** *For all $\theta$ and $m\geq 0$,*

$$\Pi_n(k_\theta)z^m=e^{i(n+2m)\theta}z^m.$$

*Hence the $K$-types of $\Pi_n$ are $n,n+2,n+4,\dots$, each with multiplicity one, and the $K$-finite vectors are the polynomials.*

*Proof.* By the formula (23.2), the element $h=\varphi(k_\theta)$ has $h^{-1}=\mathrm{diag}(e^{i\theta},e^{-i\theta})$. So $h^{-1}\cdot z=e^{2i\theta}z$ and $j(h^{-1},z)=e^{-i\theta}$. Now (23.5) gives $e^{in\theta}(e^{2i\theta}z)^m$.

For an integer $k$ define

$$P_k f=\frac{1}{2\pi}\int_0^{2\pi}e^{-ik\theta}\,\Pi_n(k_\theta)f\,d\theta,$$

a Riemann integral of a continuous $\mathcal{A}_n$-valued function. Then $P_k e_m=e_m$ if $k=n+2m$, and $P_k e_m=0$ otherwise. Being bounded, $P_k$ is the orthogonal projection onto $\mathbb{C}e_m$, or zero. Its range is the space of vectors of $K$-type $k$.

If $v$ is $K$-finite, its orbit spans a finite-dimensional space $U$. Every $P_k v$ lies in $U$, as a limit of Riemann sums there. The nonzero $P_k v$ are orthogonal, so only finitely many occur. The basis expansion of $v$ is therefore a polynomial. Conversely, every polynomial is a finite sum of $K$-types. ∎

**Lemma 23.7.** *Polynomials are smooth vectors for $\Pi_n$. For $Y=\begin{pmatrix}p&q\\r&s\end{pmatrix}\in\mathfrak{su}(1,1)$ and $a$*

*polynomial $f$,*

$$d\Pi_n(Y)f = n(rz+s)f + \big(rz^2 + (s-p)z - q\big)f'. \tag{23.6}$$

*Proof.* By the lower bound for $|j(h,z)|$ in Lemma 23.1(a), $(h,z) \mapsto (\Pi_n(h)f)(z)$ is smooth on $\mathrm{SU}(1,1) \times \overline{D}$. By Taylor's theorem on compact sets, difference quotients in $h$ of every order converge uniformly on $\overline{D}$. By the supremum bound in Lemma 23.3, they converge in $\mathcal{A}_n$. So $f$ is a smooth vector in the sense of Lecture 21. Moreover $d\Pi_n(Y)f$ is the pointwise $t$-derivative at 0 of $(\Pi_n(h_t)f)(z)$, where $h_t = \exp(tY)$. Since $h_t^{-1} = I - tY + O(t^2)$, uniformly in $z \in \overline{D}$,

$$j(h_t^{-1}, z) = 1 - t(rz+s) + O(t^2), \qquad h_t^{-1} \cdot z = \frac{(1-tp)z - tq}{1 - t(rz+s)} = z + t\big(rz^2 + (s-p)z - q\big) + O(t^2).$$

So $j(h_t^{-1}, z)^{-n} = 1 + nt(rz+s) + O(t^2)$, and the product and chain rules give (23.6). ∎

**Proposition 23.8.** *Extend $d\Pi_n$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. Then*

$$\begin{aligned} d\Pi_n(H')z^m &= (n+2m)z^m, \\ d\Pi_n(E)z^m &= (n+m)z^{m+1}, \\ d\Pi_n(F)z^m &= -m\,z^{m-1} \quad (m \geq 1), \qquad d\Pi_n(F)1 = 0. \end{aligned}$$

*The Casimir element $\Omega = H'^2 + 2EF + 2FE$ acts on every polynomial by $n(n-2)$.*

*Proof.* The right side of (23.6) is complex linear in $(p,q,r,s)$. So it also computes $d\Pi_n$ on the matrices in (23.3). For $H'$ we have $p=-1$, $s=1$, giving $nf + 2zf'$. For $E$ we have $r=1$, giving $nzf + z^2f'$. For $F$ we have $q=1$, giving $-f'$. Apply these to $z^m$. Since $[E,F] = H'$, we have $\Omega = H'^2 + 2H' + 4FE$. On $z^m$ its eigenvalue is

$$(n+2m)^2 + 2(n+2m) - 4(n+m)(m+1) = n^2 - 2n.$$ ∎

For a unitary representation $\Pi$ and $X \in \mathfrak{g}$, the operator $d\Pi(X)$ is skew-symmetric on smooth vectors. Extend $d\Pi$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. The matrices $H_0$, $S$ and $W$ are real, and the inner product is antilinear in the first slot. Moving $d\Pi(E)$ to the second slot therefore negates the real generators $H_0$ and $S$. It also turns the coefficient $i$ of $S$ into $-i$, which produces $-d\Pi(F)$. For $H' = -iW$ the two sign changes cancel. Hence smooth vectors $u$ and $v$ satisfy

$$\langle d\Pi(E)u, v\rangle = -\langle u, d\Pi(F)v\rangle, \qquad \langle d\Pi(H')u, v\rangle = \langle u, d\Pi(H')v\rangle. \tag{23.7}$$

In the model, $u = z^m$ and $v = z^{m+1}$ give $(n+m)c_{m+1}$ on the left and $(m+1)c_m$ on the right. These agree by the formula for $c_m$ in Lemma 23.3. Conversely, distinct $K$-types are orthogonal, so (23.7) forces $c_{m+1}/c_m = (m+1)/(n+m)$. Up to a scalar, the weighted norm is the only such inner product on polynomials.

**Theorem 23.9.** *$\Pi_n$ is irreducible for every integer $n \geq 2$.*

*Proof.* Let $V \neq 0$ be a closed invariant subspace and $0 \neq v \in V$. Each $P_k$ maps $V$ into $V$, since its Riemann sums do and $V$ is closed. As $P_{n+2m}v = \langle e_m, v\rangle e_m$, some $e_m$ lies in $V$. For $X \in \mathfrak{g}$, $d\Pi_n(X)e_m$ is the norm limit of $t^{-1}(\Pi_n(\exp tX)e_m - e_m) \in V$, so it lies in $V$. Hence $d\Pi_n(E)e_m$ and $d\Pi_n(F)e_m$ lie in $V$. By the operator formulas of Proposition 23.8, $E$ sends $z^m$ to $(n+m)z^{m+1}$ with $n+m \neq 0$. For $m \geq 1$, $F$ sends $z^m$ to $-mz^{m-1}$ with $m \neq 0$. So every $z^k$ lies in $V$. Polynomials are dense, so $V = \mathcal{A}_n$. ∎

An irreducible unitary representation belongs to the *discrete series* if its matrix coefficients are square integrable on $G$. The family $\Pi_n$, $n \geq 2$, is the holomorphic discrete series in our labelling. Its square integrability is cited from Knapp's §VI.1; Exercise 23.6 checks the coefficient of $e_0$. The end of § 23.4 explains the word discrete.

## 23.3 Descent conditions

**Proposition 23.10.** *Let $\Pi$ be a strongly continuous representation of $G$ on a Hilbert space. Let $v \neq 0$ have $K$-type $k$.*

*(a) $k \in \mathbb{Z}$. In particular, for $\lambda \notin \mathbb{Z}$ no such $\Pi$ has a lowest-weight module with weights $\lambda, \lambda+2, \dots$ as its $K$-finite part.*

*(b) $\Pi(-I)v = (-1)^k v$.*

*(c) $\Pi_n(-I) = (-1)^n \mathrm{id}$. Hence $\Pi_n$ factors through $\mathrm{PSL}(2,\mathbb{R}) = G/\{\pm I\}$ if and only if $n$ is even.*

*Proof.* (a) Since $k_{2\pi} = I$, we have $v = \Pi(k_{2\pi})v = e^{2\pi i k}v$. As $v \neq 0$, this forces $e^{2\pi i k} = 1$, so $k \in \mathbb{Z}$. A lowest-weight $K$-finite part with weights $\lambda, \lambda+2, \dots$ contains a vector of $K$-type $\lambda$, so it requires $\lambda \in \mathbb{Z}$.

(b) Since $k_\pi = -I$, the definition of $K$-type at $\theta = \pi$ gives $\Pi(-I)v = e^{i\pi k}v$. This equals $(-1)^k v$ because $k \in \mathbb{Z}$.

(c) By the formula (23.2) for $\varphi(-I)$, $(-I)\cdot z = z$ and $j(-I,z) = -1$, so $\Pi_n(-I)f = (-1)^{-n}f$. The quotient map $G \to \mathrm{PSL}(2,\mathbb{R})$ is onto with kernel $\{\pm I\}$. So $\Pi_n$ factors exactly when $\Pi_n(-I) = \mathrm{id}$. ∎

The value of $\Pi_n(-I)$ in Proposition 23.10(c) agrees with the sign rule of Proposition 23.10(b). Indeed, every $K$-type $n+2m$ of $\Pi_n$ has the parity of $n$. The integrality argument of Proposition 23.10(a) uses the period $2\pi$ of $K$ in $G$. On the universal covering group the lift of $K$ is a copy of $\mathbb{R}$, so (a) gives no condition there. We do not study that group.

**Remark 23.11.** The ratio forced by (23.7) still makes sense for $n = 1$. It gives $c_m = 1$, so the norm is $\sum |a_m|^2$. No positive measure $\mu$ on $D$ gives this norm by area integration. It would satisfy $\int_D 1\, d\mu = \int_D |z|^2 d\mu = 1$, although $1 - |z|^2$ is strictly positive on $D$. This is impossible. So lowest weight 1 lies outside the family $\mathcal{A}_n$. It appears below as a limit of discrete series.

## 23.4 Comparison and the unitary picture

Let $V$ be a $(\mathfrak{g}, K)$-module of $G$ (Lecture 21) on which $\Omega$ acts by a scalar $c$. Write $v_k$ for a vector of $K$-type $k$. Since $\Omega = H'^2 + 2H' + 4FE = H'^2 - 2H' + 4EF$,

$$4FEv_k = \big(c - k(k+2)\big)v_k, \qquad 4EFv_k = \big(c - k(k-2)\big)v_k. \tag{23.8}$$

**Example 23.12.** Fix $n \geq 2$ and $c = n(n-2) = (n-1)^2 - 1$. By (23.8), $FEv_k = 0$ exactly when $k \in \{-n, n-2\}$.

(i) In $\Pi_n$ the $K$-types are $n, n+2, \dots$. $F$ kills $z^0$, and $E$ is injective.

(ii) Put $d = n-2$ and replace $n$ by $-d$ in (23.5). Call the resulting operator $\Pi_{-d}(h)$. It sends $f$ to $j(h^{-1}, z)^d f(h^{-1} \cdot z)$ and preserves the polynomials of degree at most $d$. The computation of (23.6) gives $d\Pi(E)z^m = (m-d)z^{m+1}$, which kills $z^d$. The $K$-types are $-d, \dots, d$. On the lowest vector $v_{-d} = z^0$, we have $Fv_{-d} = 0$. Thus (23.8) gives $c = (-d)(-d-2) = d(d+2) = n(n-2)$. Exercise 23.3 gives the polynomial matrices.

(iii) The principal series of Lecture 22 has every integer of one parity as a $K$-type, once each. Take the principal series module $V_{\varepsilon,\nu}$ of Lecture 22 with $\nu = n-1$, and with $\varepsilon$ chosen so that its $K$-types have the parity of $n$. By Lecture 22, Exercise 22.4, $\Omega$ acts on it by $\nu^2 - 1 = n(n-2)$. At $k = n-2$, (23.8) says that the coefficient of $E\colon v_{n-2} \mapsto v_n$ times that of $F\colon v_n \mapsto v_{n-2}$ vanishes. The same holds between $-n$ and $2-n$.
The ladder of Lecture 22 says which arrow vanishes. At $\nu = n-1$ the arrows $E$ from $-n$ and $F$ from $n$ vanish. So the half-ladders $k \ge n$ and $k \le -n$ are submodules, and the finite segment $2-n \le k \le n-2$ is what remains when both are removed. At $\nu = -(n-1)$ the arrows $E$ from $n-2$ and $F$ from $2-n$ vanish, so the finite segment is a submodule. In both cases the three pieces have the $K$-types and Casimir value of $\Pi_n^-$, of the finite-dimensional module $\Pi_{-d}$ with $d = n-2$, and of $\Pi_n$.

We now identify the piece $k \ge n$ with $\Pi_n$. In this lecture $e_m$ is the basis of $\mathcal{A}_n$, so we write $\mathsf{e}_k(\theta) = e^{ik\theta}$ for the Fourier vectors of Lecture 22. For $n \ge 1$ and $f$ holomorphic on a neighbourhood of $\overline{D}$, put

$$(U_n f)(\theta) = e^{in\theta} f(e^{2i\theta}), \qquad \text{so that} \qquad U_n z^m = \mathsf{e}_{n+2m}.$$

This is the map $U_n$ of Lecture 22, with the same formula. Formula (23.5) also makes sense for $n = 1$ on such functions.

**Proposition 23.13.** *Let $n \ge 1$ and let $\varepsilon \equiv n \pmod 2$. For a polynomial $f$ and $g \in G$, the function $\Pi_n(g)f$ is holomorphic on a neighbourhood of $\overline{D}$, and*

$$U_n \Pi_n(g) f = \Pi_{\varepsilon,n-1}(g) U_n f.$$

*Proof.* Put $h = \varphi(g)^{-1} = \gamma g^{-1} \gamma^{-1}$. By (23.5), $(\Pi_n(g)f)(z) = j(h,z)^{-n} f(h \cdot z)$. By Lemma 23.1(a), $j(h, \cdot)$ has no zero on $\overline{D}$, and its only possible zero lies outside. So $\Pi_n(g)f$ is holomorphic near $\overline{D}$.

Let $z = e^{2i\theta}$ and $g^{-1}u_\theta = r u_\tau$, with $u_\theta = (\cos\theta, \sin\theta)^T$ as in Lecture 22. Then $\gamma u_\theta = e^{-i\theta}(z, 1)^T$. Applying $\gamma g^{-1} = h\gamma$ to $u_\theta$ gives $h \cdot z = e^{2i\tau}$ and $j(h,z) = r e^{i(\theta-\tau)}$, as in the half-ladder formula of Lecture 22. Hence

$$e^{in\theta} j(h,z)^{-n} f(h \cdot z) = r^{-n} e^{in\tau} f(e^{2i\tau}) = (\Pi_{\varepsilon,n-1}(g) U_n f)(\theta). \qquad \blacksquare$$

This identity uses only matrices, so the opposite sign of $k_\theta$ in this lecture does not enter. It says that the polynomials of $\Pi_n$ are the piece $k \ge n$ of $V_{\varepsilon,n-1}$. The ladders agree too: for $\nu = n-1$ and $k = n+2m$ we have $\frac{1}{2}(\nu+1+k) = n+m$ and $\frac{1}{2}(\nu+1-k) = -m$, which are the coefficients of Proposition 23.8. The boundary point $z = e^{2i\theta}$ is the line $\mathbb{R}u_\theta$ of Lecture 22.

Only the norm differs. The squared disk norm of $z^m$ is $c_m$, and the squared circle norm of $\mathsf{e}_{n+2m}$ is 1. So for $n \ge 2$ the map $U_n$ is not an isometry. Also a general vector of $\mathcal{A}_n$ need not have boundary values, so we do not define $U_n$ on all of $\mathcal{A}_n$. At $n = 1$ we have $c_m = 1$, and the two norms agree.

**Corollary 23.14.** *Let $\mathcal{A}_1$ be the completion of the polynomials in the norm $\|\sum a_m z^m\|^2 = \sum |a_m|^2$. Then $U_1$ extends to a unitary map from $\mathcal{A}_1$ onto the closed span $H_1^+$ of the vectors $\mathtt{e}_1, \mathtt{e}_3, \dots$. The operators $\Pi_1(g) = U_1^{-1}\Pi_{1,0}(g)U_1$ form an irreducible strongly continuous unitary representation of $G$ on $\mathcal{A}_1$, and on polynomials they are given by (23.5) with $n = 1$. Put $\Pi_1^-(g) = \Pi_1(\delta g \delta^{-1})$. Then $\Pi_{1,0}$ is the orthogonal sum of a copy of $\Pi_1$ on $H_1^+$ and a copy of $\Pi_1^-$ on the closed span of $\mathtt{e}_{-1}, \mathtt{e}_{-3}, \dots$.*

*Proof.* By Parseval, $\|U_1 f\|^2 = \sum |a_m|^2$, and the vectors $\mathtt{e}_{1+2m}$ are an orthonormal basis of $H_1^+$. So $U_1$ extends to a unitary map onto $H_1^+$. The representation $\Pi_{1,0}$ is unitary and strongly continuous, since $\nu = 0$ is imaginary. By the irreducibility theorem of Lecture 22, $H_1^+$ is invariant and irreducible. These properties pass to $\Pi_1$ through $U_1$. A function holomorphic near $\overline{D}$ has a Taylor series that converges in $\mathcal{A}_1$, so the extended $U_1$ agrees on it with the formula above. Hence, on polynomials, Proposition 23.13 gives the stated formula for $\Pi_1(g)$.

For the second half, put $(R\varphi)(\theta) = \varphi(-\theta)$. This is unitary and carries $H_1^+$ onto the negative half. Since $\delta u_\theta = u_{-\theta}$, the circle formula of Lecture 22 gives $\Pi_{1,0}(\delta g \delta^{-1}) = R\,\Pi_{1,0}(g)R$. Hence $RU_1\Pi_1^-(g) = \Pi_{1,0}(g)RU_1$. ∎

We now compare the sign in the two real forms of $\mathfrak{sl}(2,\mathbb{C})$. In a unitary representation of SU(2) the same triple has, in physics notation, $H' = 2J_3$, $E = J_+$ and $F = J_-$, and now $\langle Eu, v\rangle = \langle u, Fv\rangle$. Then this sign and (23.8) give $\|Ev_k\|^2 = \frac{1}{4}\big(c - k(k+2)\big)\|v_k\|^2$. For $c = 4j(j+1)$ and $k = 2m$, this is $j(j+1) - m(m+1)$ times $\|v_k\|^2$. For large $k$ the right side is negative, so the ladder must stop. For $\mathrm{SL}(2,\mathbb{R})$ the relation (23.7) has a minus sign, and the sign of the norm formula is reversed. So a ladder need not stop.

**Proposition 23.15.** *Let $V$ be an irreducible $(\mathfrak{g}, K)$-module of $G$. Assume each $K$-type has multiplicity at most one and $\Omega$ acts by $c$. Suppose $V$ has a positive definite inner product satisfying (23.7) for the module action. If $k$ and $k+2$ are $K$-types, then*

$$\|Ev_k\|^2 = \tfrac{1}{4}\big(k(k+2) - c\big)\|v_k\|^2 > 0. \tag{23.9}$$

*Consequently:*

*(i) if the $K$-types are $\lambda, \lambda+2, \lambda+4, \dots$, then $\lambda \geq 1$ is an integer and $c = \lambda(\lambda-2)$;*

*(ii) if the $K$-types are $2\mathbb{Z}$, then $c < 0$;*

*(iii) if the $K$-types are $2\mathbb{Z}+1$, then $c < -1$.*

*Proof.* By (23.7) and (23.8), $\|Ev_k\|^2 = -\langle v_k, FEv_k\rangle = \frac{1}{4}(k(k+2) - c)\|v_k\|^2$. Since $\|Ev_k\|^2$ and $\|v_k\|^2$ are real and $v_k \neq 0$, this identity shows that $c$ is real.

Next we show that $Ev_k \neq 0$, arguing by contradiction. Suppose $Ev_k = 0$. For each parity, the sum of the $K$-type spaces of that parity is a submodule. By irreducibility, all $K$-types have the parity of $k$. Since $E$ raises and $F$ lowers $K$-types by 2, the span of the $v_j$ with $j \leq k$ is invariant. It is nonzero and misses $v_{k+2}$, which contradicts irreducibility. So (23.9) is strict.

For the lowest-weight pattern (i), $Fv_\lambda = 0$, so (23.8) gives $c = \lambda(\lambda-2)$. Then (23.9) at $k = \lambda$ reads $\|Ev_\lambda\|^2 = \lambda\|v_\lambda\|^2$, so $\lambda > 0$. The circle action on the $(\mathfrak{g}, K)$-module satisfies $k_{2\pi} = I$. Thus $v_\lambda = e^{2\pi i\lambda}v_\lambda$, which makes $\lambda$ an integer. For the two-sided patterns (ii) and (iii), write $k(k+2) = (k+1)^2 - 1$. Its minimum is 0 over even $k$ and $-1$ over odd $k$. ∎

Figure 23.2 shows formula (23.9). At $c = 8$ the dots on the line are $k = 2$ and $k = -4$, where the step vanishes. Between them the even dots $k = -2$ and $k = 0$ are the steps of the finite-dimensional module of $K$-types $-2, 0, 2$. For $\mathrm{SL}(2,\mathbb{R})$ these steps have negative squared norm. For $\mathrm{SU}(2)$ they are exactly the steps with positive squared norm.

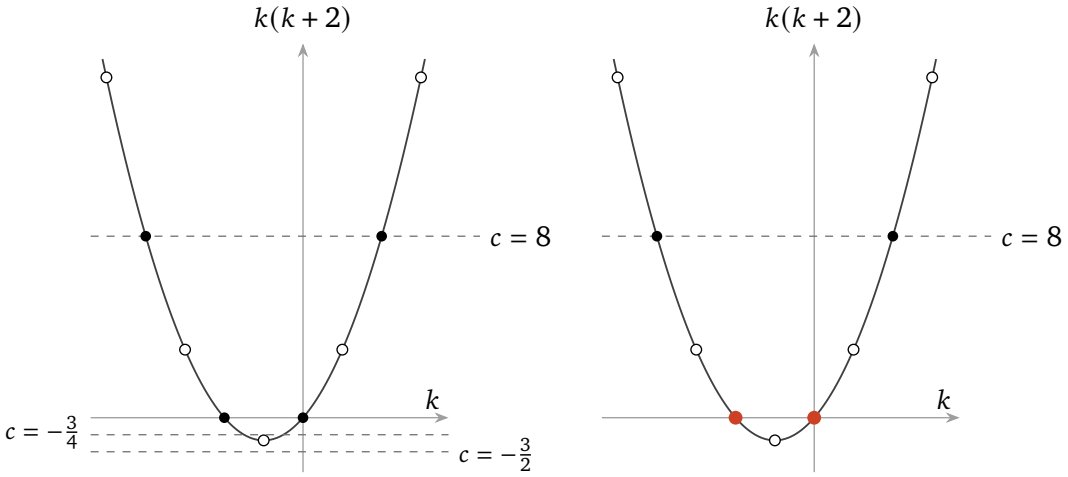



Figure 23.2: Formula (23.9) for $\mathrm{SL}(2,\mathbb{R})$ in the first panel and its $\mathrm{SU}(2)$ analogue in the second. Filled dots are even $k$ and hollow dots are odd $k$. In the first panel a nonzero step from $k$ to $k+2$ needs its dot strictly above the line $y = c$; a dot on the line is a zero step. In the second panel the step needs its dot strictly below the line, and the coloured dots are the steps of the module with $K$-types $-2, 0, 2$.

Twisting by $\delta$ from Remark 23.5 interchanges $E, F$, sends $H'$ to $-H'$ and preserves (23.7). Thus the lowest-weight case (i) also gives $c = \lambda(\lambda - 2)$ with integer $\lambda \geq 1$ for highest-weight patterns $-\lambda, -\lambda - 2, \ldots$.

**Theorem 23.16** (Bargmann; cited). *Every irreducible unitary representation of $G$ is equivalent to exactly one entry of the following list. Each entry is irreducible and unitary, and is determined by its $K$-types and its Casimir value $c$. Here $\Pi_1$ and $\Pi_1^-$ are the representations of Corollary 23.14, whose unitarity and irreducibility are proved there.*

| *Family* | *$K$-types* | $c$ | $\nu$ |
|---|---|---|---|
| *trivial* | $\{0\}$ | $0$ | $\pm 1$ |
| *even principal series* | $2\mathbb{Z}$ | $c \leq -1$ | $i\mathbb{R}$ |
| *odd principal series* | $2\mathbb{Z}+1$ | $c < -1$ | $i\mathbb{R} \setminus \{0\}$ |
| *complementary series* | $2\mathbb{Z}$ | $-1 < c < 0$ | $(-1,1) \setminus \{0\}$ |
| $\Pi_n$, $n \geq 1$ | $n, n+2, \ldots$ | $n(n-2)$ | $\pm(n-1)$ |
| $\Pi_n^-$, $n \geq 1$ | $-n, -n-2, \ldots$ | $n(n-2)$ | $\pm(n-1)$ |

*For $n \geq 2$ the last two rows are the discrete series. For $n = 1$ they are the limits of discrete series, realised with the norm of Remark 23.11. Their sum is the odd principal series at $c = -1$.*

The last column gives the parameters of Lecture 22 with $c = \nu^2 - 1$. In the principal and complementary rows the entry has the module $V_{\varepsilon,\nu}$, with $\varepsilon = 0$ in the even rows and $\varepsilon = 1$ in the odd row, and $\nu$ and $-\nu$ give the same module. In the other rows the entry is only one

piece of a reducible module $V_{\varepsilon,\nu}$. So a value of $\nu$ can carry several entries of the list.

Proposition 23.15 proves the necessity of the $c$-ranges for these $K$-type patterns. Call a $(\mathfrak{g}, K)$-module *even* if its $K$-types are $2\mathbb{Z}$, and *odd* if they are $2\mathbb{Z}+1$. An irreducible even module with a positive definite inner product satisfying (23.7) needs $c < 0$. The part $c \le -1$ is the unitary principal series of Lecture 22. The interval $-1 < c < 0$ is the complementary series. The inner product of the complementary series is fixed by (23.9), not by the $L^2$ norm of Lecture 22 (Exercise 23.5). An irreducible odd module with such an inner product needs $c < -1$, so there is no odd complementary series. At $c = -1$ the odd principal series module has $FEv_{-1} = 0$ and splits.

Figure 23.3 sets the list against $c$. We use $c$ and not $\nu$ because of the last remark on the table. At $c = -1$ the odd principal series splits into $\Pi_1$ and $\Pi_1^-$. At $c = 0$ the representations $\Pi_2$, $\Pi_2^-$ and the trivial representation are distinct.

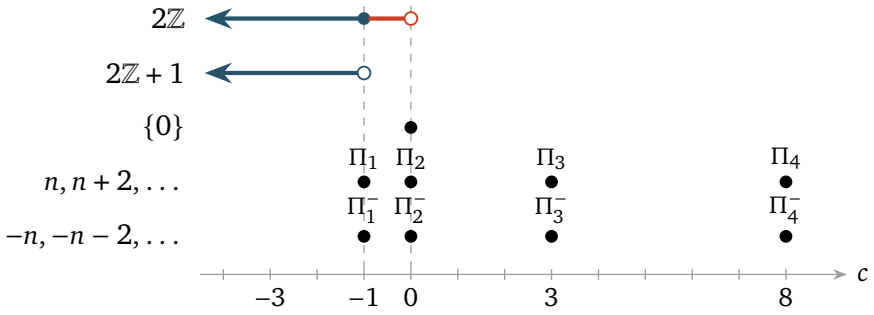



Figure 23.3: Bargmann's list against the Casimir value $c$, with one row for each pattern of $K$-types. In the top row the long segment $c \le -1$ is the even principal series, and the short segment $-1 < c < 0$ is the complementary series. The second row is the odd principal series, the third is the trivial representation, and the last two rows show the first few $\Pi_n$ and $\Pi_n^-$.

The cited parts are the converse and the exhaustion. They use two results of Harish-Chandra (compare Lecture 21). First, every irreducible unitary representation is admissible. Its $K$-finite vectors form an irreducible $(\mathfrak{g}, K)$-module with multiplicity-one $K$-types on which $\Omega$ is scalar. Second, each module above with an invariant positive definite Hermitian form integrates to an irreducible unitary representation of $G$. Etingof's MIT 18.757 notes treat admissibility in Lecture 5 and unitary globalisation in Theorem 7.5.

We can now explain the word discrete. For a compact group, the Plancherel theorem of Lecture 11 writes the squared norm of a function as a sum over the irreducible representations. For $G$ the corresponding formula has a sum and integrals. Let $h$ be smooth with compact support on $G$, and put $\Pi(h) = \int_G h(g)\Pi(g)\,dg$. For a suitable normalisation of Haar measure,

$$h(I) = \int_{\mathbb{R}} \operatorname{tr} \Pi_{0,i\nu}(h)\, \nu \tanh \frac{\pi\nu}{2}\, d\nu + \int_{\mathbb{R}} \operatorname{tr} \Pi_{1,i\nu}(h)\, \nu \coth \frac{\pi\nu}{2}\, d\nu + \sum_{n \ge 2} 4(n-1) \operatorname{tr}\bigl(\Pi_n(h) + \Pi_n^-(h)\bigr).$$

We cite this inversion formula from Knapp, §II.7, formula (2.25), and we do not use it. Knapp proves it as Theorem 11.6 in §XI.3 and credits it to Harish-Chandra (1952). The Plancherel formula for $G$ follows from it. The discrete series enters through a discrete sum, and the

unitary principal series enters through integrals over $\nu = i\nu$. Neither the complementary series nor $\Pi_1^{\pm}$ appears. The number $n-1$ is the one in Exercise 23.6, where the coefficient of $e_0$ has squared norm $C/(n-1)$.

Finally we return to finite dimensions. The next statement does not use Bargmann's theorem, and it needs no irreducibility hypothesis.

**Corollary 23.17.** *Every finite-dimensional continuous unitary representation $\Pi$ of $G$ is trivial.*

*Proof.* By the theorem of Lecture 3 on continuous homomorphisms, $\Pi$ is smooth. The complex-linear extension of $d\Pi$ makes the space $V$ a finite-dimensional module of $\mathfrak{sl}(2,\mathbb{C})$. The real element $H_0 = \operatorname{diag}(1,-1)$ is the diagonal element of the triple of Lecture 20. So complete reducibility and the highest-weight calculation of the prerequisite Lie Algebras course make $d\Pi(H_0)$ diagonalisable with integer eigenvalues. Unitarity makes the same operator skew-Hermitian, with imaginary eigenvalues. Therefore $d\Pi(H_0) = 0$.

The kernel of the complexified differential is an ideal of $\mathfrak{sl}(2,\mathbb{C})$ containing the nonzero element $H_0$. Since $\mathfrak{sl}(2,\mathbb{C})$ is simple, this kernel is everything, so $d\Pi = 0$. Since $G$ is connected, it is generated by exponentials (Lecture 2), and $\Pi(\exp X) = \exp d\Pi(X) = I$. Hence $\Pi$ is trivial. ∎

So a nontrivial unitary representation of $G$ is infinite-dimensional. Exercise 23.3 checks this directly for the finite-dimensional modules $\Pi_{-d}$ of Example 23.12(ii).

## 23.5 Self-checks

1. For $n = 3$, compute $c_1$ and $c_2$. Check (23.7) for $u = z$ and $v = z^2$.

   **Answer.** $c_1 = 1!\,2!/3! = 1/3$ and $c_2 = 2!\,2!/4! = 1/6$. The left side is $(3+1)c_2 = 2/3$. The right side is $-\langle z, -2z\rangle = 2c_1 = 2/3$.

2. Does $\Pi_5$ factor through $\mathrm{PSL}(2,\mathbb{R})$? Compute $\Pi_5(k_{\pi/2})z^m$.

   **Answer.** No, since $\Pi_5(-I) = -\mathrm{id}$. By the $K$-type formula of Proposition 23.6, $\Pi_5(k_{\pi/2})z^m = i^{5+2m}z^m = (-1)^m\, i\, z^m$.

3. Is there an irreducible unitary representation with $K$-types $1, 3, 5, \ldots$ and $c = -\frac{1}{2}$?

   **Answer.** No. By the cited Harish-Chandra facts after Theorem 23.16, the $K$-finite vectors of such a representation satisfy the hypotheses of Proposition 23.15. Its lowest-weight case (i) shows that the lowest $K$-type 1 forces $c = 1\cdot(1-2) = -1$.

## 23.6 Summary

For each integer $n \geq 2$, formula (23.5) defines an irreducible unitary representation $\Pi_n$ on $\mathcal{A}_n$. Its $K$-types are $n, n+2, \ldots$, each once. The operator $E$ raises the $K$-type by 2 and $F$ lowers it by 2. The Casimir element $\Omega$ acts by $n(n-2)$. The relation $k_{2\pi} = I$ forces integral $K$-types. The representation $\Pi_n$ descends to $\mathrm{PSL}(2,\mathbb{R})$ exactly for even $n$. In an irreducible unitary module with $K$-types $\lambda, \lambda+2, \ldots$, the lowest weight is an integer $\lambda \geq 1$, and $\Omega$ acts

by $\lambda(\lambda-2)$. Positivity of $\|Ev_k\|^2$ explains the shape of Bargmann's cited list, including the complementary series.

The map $U_n$ carries the polynomials of $\Pi_n$ onto the half-ladder $k \geq n$ of the principal series at $\nu = n-1$, and only the norm differs. For $n=1$ the norms agree, and this constructs $\Pi_1$ as a summand of the odd principal series at $\nu = 0$. Every finite-dimensional continuous unitary representation of $G$ is trivial. We also cited the inversion formula for $G$, whose discrete sum runs over the discrete series.

## 23.7 Reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001 edition. §VI.1 treats the holomorphic discrete series of $\mathrm{SU}(1,1)$. §§II.5–II.6 treat the unitary representations of $\mathrm{SL}(2,\mathbb{R})$. §II.7 states the inversion formula cited above, and §XI.3 proves it.
- MIT 18.112, *Functions of a Complex Variable*, Fall 2008: Cauchy's formula, power series and the Cauchy–Riemann equations.
- P. Etingof, *Representations of Lie Groups*, MIT 18.757 lecture notes, Fall 2023. Lecture 5 treats admissibility, and Theorem 7.5 is the unitary globalisation theorem.
- Harish-Chandra, Plancherel formula for the $2\times 2$ real unimodular group, *Proc. Nat. Acad. Sci. USA* 38 (1952), 337–342.
- V. Bargmann, Irreducible unitary representations of the Lorentz group, *Ann. of Math.* 48 (1947), 568–640, for the original classification.